

Finding the Best Neighborhood to Live: Heilmeier Questions

According to Census data, relocation in America is at its lowest rate on record (Tavernise, 2019). This is concerning because the lack of people moving for work could negatively impact the economy (Kopf, 2018). Still, trying to find the right place to live can be difficult. We hope that our project can reverse the downward mobility trends by helping people find a new neighborhood that meets their needs.

Today, a few popular websites help people find the “best places to live”. Livability and Teleport provide a profile of cities and a breakdown of their basic statistics. Niche and AreaVibes offer further data on neighborhoods. “Livability” scores are calculated based on several categories, such as cost, crime, etc. These individual groupings also receive a rating, and all scores contribute to a locale’s ranking in comparison to nearby areas. Finally, the data presentations are static and arranged from top to bottom.

Regrettably, there are several limits to the current practice. First, none of these sites offer a list of neighborhoods similar to the one a user is currently reviewing. Next, these sites simply offer users the ability to search for different locations. While searches are filterable, there are no personalized recommendations for the user. Moreover, these sites fail to offer any information about local grocery stores or other essential businesses in an area, which is important when deciding on a new residence. Lastly, aside from home sale maps, none of the sites offer an interactive visualization.

So, our group project focuses on making a few improvements that are not available. To start, we plan to improve the underlying dataset by including information on essential businesses. Then, we will provide a group of neighborhoods around the US that are similar to the one a user provides. If possible, we will also provide a list of top neighborhoods a user should consider. To be more precise, our project will determine the neighborhood of a user inputted address, and then it will utilize the improved dataset by performing a clustering algorithm that will group this neighborhood with similar ones from across the US. If time permits, we will include a recommendation algorithm, which will predict how strongly neighborhoods are connected to user inputs. Finally, the results will be presented in an interactive visual.

Our project will be successful because it will greatly ease the search for a new locale. Our goal is to provide people with personalized data about neighborhoods to help them make informative decisions, which we believe will encourage more mobility. The measures of success for our algorithms will be based on volume, variety, and velocity. The algorithms should handle a large dataset with high dimensionality, have a practical run time, and generate stable and reasonable results. If our project were to launch, then success could be measured through user surveys, focus groups, and engagement metrics.

The risks of the project are associated with data quality and integrity, how we process the data, and the validity of the results. However, we believe the payoffs outweigh the risks. Users will feel comfortable about choosing a suitable neighborhood because the time and stress of searching will be reduced. Above all, our tool will give people the confidence to relocate when they might have been hesitant to before.

Monetary costs will not be associated with this project, only development time. The tools we will use are free. Currently, we plan to utilize SQLite and Python. Our visualization tool is yet to be chosen.

By the “midterm exam” check, our group plans to have: all data retrieved, organized, and cleaned; the clustering algorithm ready for testing; and a “proof of concept” for the recommendation algorithm and interactive visualization. A functional prototype will be ready in 6 weeks. At the “final exam” check, we will submit an interactive visualization that utilizes reasonable results produced by a stable and efficient clustering algorithm. If time permits, then we will also include the output from a recommendation algorithm. Finally, our progress will be measured by the completion of the tasks listed in the next section.

Plan of Activities and Time Estimates, Per Group Member

Thus far, all team members have provided similar effort. We have all researched potential data sources, interacted with the APIs, and equally contributed to the project proposal. Our project’s plan of activities, time estimates, and team member assignments are: data retrieval, organization, and cleaning (2 weeks: all team members); model construction, validation, and testing (2 weeks: Janicki, Meyer, Bell, & Smootha); and development of the interactive visualization (2 weeks: Dassanayake & Bodie).

Literature Survey

First, the GARM chapter provided details on census tracts and their construction (U.S. Census Bureau, 1994). They are often used as proxies for neighborhoods, which our group will exploit. This reference

was strictly an informative source, so there were no shortcomings to consider. Next, several papers looked at neighborhood factors people find important, which our group will consider. Okulicz-Kozaryn (2011) compared livability and satisfaction of locations and finds only a .36 correlation. Though, there may have been a bias towards subjective metrics. Our group intends to treat all attributes equally. Ayub et al. (2020) examined housing indicators to understand the resilience of housing markets, but the focus was strictly on urban markets. For our dataset, we will collect census tracts from every type of locale. He and Xia (2020) determined rental effects for a housing market are largely based on area amenities. Yet, their analysis only focused on Barcelona, while we will highlight the entire US. Azadeh and Moghaddam (2012) showed economic and government factors can predict fluctuations in housing markets, but they did not consider other factors. For our dataset, we will try to incorporate as many neighborhood attributes as possible. Ashkexari-Toussi et al. (2019) assessed city sentiment in geotagged photos, but they used data from 1999. The data we retrieve will be the most recent available.

A key component of our project will be to utilize a cluster model to group similar neighborhoods. Fahad et al. (2014) evaluates well-known single clustering algorithms. However, there is not a single algorithm that performs best on all datasets. Cluster ensembles can overcome this issue by combining different clustering outputs into one consensus result, and they are surveyed in Boongoen and Iam-On (2018). Though, since our final dataset may contain numerous attributes, then a multi-clustering (MC) solution that produces multiple structures on the data may be warranted. Hu and Pei (2017) provided a review of the different subspace MC approaches, which are useful for high-dimensional data. Though, only subspace MC solutions were surveyed, so more research will be needed on other MC methods.

In addition, we reviewed ranking and recommendation models as other possible approaches. Kiseleva et al. (2015) showed that a Naïve Bayes ranking on features for location recommendations performed better than other rankings, but historical data was not leveraged. Our intent is to allow users to input the locations they have lived. Liu and Yang (2015) improved VSRank recommendation by focusing on negative similarity and people's preferences. Unfortunately, geography was not considered, but this will be one of our key factors. Kim et al. (2020) improved collaborative filtering-based recommendations with sentiment analysis, but it could be improved by specialized sentiment rather than dictionary-based analysis. Su et al. (2020) created a vector of multiple similarities, which our project could utilize. Though, only MAE and RMSE were used to measure accuracy. We will use additional metrics for our evaluations. Bobadilla et al. (2013) reviewed several recommendation algorithms, but none of them were as effective with sparse data or cold starts. To overcome this problem, we could consider hybrid methods. Renjith et al. (2020). examined personalized recommender systems on multiple criteria, which we aim to employ. Yet, their focus was exclusively on travel locations, while we intend to have a more granular dataset.

Moreover, Xiaoyu et al. (2017), Al-Ghossein et al. (2018), and Wasid and Ali (2018) all employed a two-stage architecture. The first paper used matching and ranking on geotagged photos to recommend attractions, yet there was a large assumption about people's photo habits. For our project, we will not rely on assumptions about human behavior. The second paper recommended hotels using clustering and ranking. Though, there may have been issues with the contextual data. Our dataset will consist exclusively of numerical and categorical data. The third paper used a two-stage approach to cluster then produce recommendations for movies. Unfortunately, the distance measure they employed was used specifically for their dataset, while we will try to produce a more generalized model. Based on these three papers, we may choose a two-stage approach of clustering and recommendation for our final model.

Finally, we looked at how various models were applied to geographical problems, which is the basis of our project. Shashank and Schuurman (2019) showed how variable selection impacts walkability across different models, but the variables could have been combined to create a composite model that was more informative. With our model, all significant factors will be considered simultaneously. Nicholson et al. (2019) used spatial regressions on census blocks to estimate disaster vulnerability. The analysis demonstrated a feasible use case of neighborhood data, but platform selection may have produced bias. Our project will include data from all census tracts to prevent bias. Sadler et al. (2019) scored areas and weighted expert opinions to improve a "healthfulness index". A potential improvement to this work would be the addition of sensitivity analysis to the variables in our model.

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